

Economics 326
Methods of Empirical Research in Economics
Lecture 14: Hypothesis testing in the multiple
regression model, Part 2

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Multiple restrictions

- ▶ Consider the model:

$$\ln(Wage_i) = \beta_0 + \beta_1 Experience_i + \beta_2 Experience_i^2 + \\ + \beta_3 PrevExperience_i + \beta_4 PrevExperience_i^2 + \beta_5 Education_i + U_i,$$

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$$H_1 : \beta_3 \neq 0 \text{ or } \beta_4 \neq 0.$$

t-statistics and multiple restrictions

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$$T_3 = \frac{\hat{\beta}_3}{SE(\hat{\beta}_3)} \text{ and } T_4 = \frac{\hat{\beta}_4}{SE(\hat{\beta}_4)}.$$

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Testing multiple exclusion restrictions

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$$Y_i = \beta_0 + \beta_1 X_{1,i} + \dots + \beta_q X_{q,i} + \beta_{q+1} X_{q+1,i} + \dots + \beta_k X_{k,i} + U_i.$$

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- ▶ Note that $q =$ **number of restrictions**;
- ▶ $n - k - 1 =$ **unrestricted residual df**, where k is the number of regressors in the **unrestricted model**.

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- ▶ Thus, we should reject H_0 when $F > c$ where c is some positive constant.

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- ▶ Similarly to the standard normal and t distributions, the F distribution has been tabulated and its critical values are available in statistical tables and statistical software such as Stata.

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- ▶ One can find the p -value by finding τ such that $F = F_{q, n-k-1, 1-\tau}$. The p -value is equal to τ .

F distribution in Stata

- ▶ To compute F critical values use

`disp invFtail( $q$ ,  $n - k - 1$ ,  $\alpha$ ).`

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- ▶ To compute p -values from F distribution use

`disp Ftail( $q$ ,  $n - k - 1$ ,  $F$ ).`

Example

- ▶ Consider the model:

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- ▶ We test

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$$\ln(Wage_i) = \beta_0 + \beta_1 Experience_i + \beta_2 Experience_i^2 + \beta_3 PrevExperience_i + \beta_4 PrevExperience_i^2 + \beta_5 Education_i + U_i.$$

- ▶ We test

$$H_0 : \beta_3 = 0, \beta_4 = 0 \text{ against } H_1 : \beta_3 \neq 0 \text{ or } \beta_4 \neq 0.$$

- ▶ $q = 2$.

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- ▶ $q = 2$.
- ▶ $\alpha = 0.05$.

Example: the unrestricted model

```
. regress lnWage Experience Experience2 PrevExperience PrevExperience2 Education
```

Source	SS	df	MS	Number of obs =	526
Model	51.3318741	5	10.2663748	F(5, 520) =	55.04
Residual	96.9978773	520	.186534379	Prob > F =	0.0000
Total	148.329751	525	.28253286	R-squared =	0.3461
				Adj R-squared =	0.3398
				Root MSE =	.4319

lnWage	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
Experience	.0471914	.0068074	6.93	0.000	.0338179	.0605649
Experience2	-.0008518	.0002472	-3.45	0.001	-.0013374	-.0003662
PrevExperi~e	.0168997	.0047331	3.57	0.000	.0076013	.0261981
PrevExperi~2	-.0003727	.0001208	-3.09	0.002	-.00061	-.0001354
Education	.0887704	.0072131	12.31	0.000	.0745999	.1029408
_cons	.2368427	.10287	2.30	0.022	.0347509	.4389346

- ▶ $SSR_{ur} = 96.9978773$.
- ▶ $n - k - 1 = 526 - 5 - 1 = 520$.

Example: the restricted model

```
. regress lnWage Experience Experience2 Education
```

Source	SS	df	MS	Number of obs	=	526
Model	48.8668114	3	16.2889371	F(3, 522)	=	85.49
Residual	99.46294	522	.190542031	Prob > F	=	0.0000
Total	148.329751	525	.28253286	R-squared	=	0.3294
				Adj R-squared	=	0.3256
				Root MSE	=	.43651

lnWage	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
Experience	.0510784	.0067937	7.52	0.000	.037732	.0644248
Experience2	-.0009941	.0002463	-4.04	0.000	-.001478	-.0005103
Education	.0852822	.0068978	12.36	0.000	.0717313	.0988331
_cons	.3688491	.0908138	4.06	0.000	.1904437	.5472544

► $SSR_r = 99.46294$.

Example: F statistic and test

- To compute the statistic:

$$F = \frac{(SSR_r - SSR_{ur}) / q}{SSR_{ur} / (n - k - 1)} = \frac{(99.46294 - 96.9978773) / 2}{96.9978773 / (526 - 5 - 1)} \approx 6.61.$$

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- The critical value:

```
. disp invFtail(2,520,0.05)  
3.0130572
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$\Rightarrow$ We reject H_0 for any $\alpha > 0.00146284$.

Example: Stata test command

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- ▶ To test that previous experience has no effect:

```
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```

- ▶ The output of this command is:

```
( 1) PrevExperience = 0  
( 2) PrevExperience2 = 0  
F( 2, 520) = 6.61  
Prob > F = 0.0015
```

Example: Stata test command

- ▶ To test that the coefficient on previous experience equal to the coefficient on experience and the coefficient on previous experience squared is zero:

```
. test (Experience==PrevExperience2) (PrevExperience2=0)
```

- ▶ The output is:

```
( 1) Experience - PrevExperience2 = 0
```

```
( 2) PrevExperience2 = 0
```

```
F( 2, 520) = 31.94
```

```
Prob > F = 0.0000
```

- ▶ Let R_{ur}^2 denote the R^2 corresponding to the unrestricted model:

$$Y_i = \beta_0 + \beta_1 X_{1,i} + \dots + \beta_q X_{q,i} + \beta_{q+1} X_{q+1,i} + \dots + \beta_k X_{k,i} + U_i.$$

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- ▶ The two models have the same dependent variable and therefore the same Total Sum-of-Squares:

$$SST = \sum_{i=1}^n (Y_i - \bar{Y})^2 = SST_{ur} = SST_r.$$

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F test: more examples

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Relationship between F and t statistics

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- ▶ The F statistic can also be used for testing a single restriction.
- ▶ In the case of a single restriction, the F test and t test lead to the **same outcome** because

$$t_{n-k-1}^2 = F_{1,n-k-1}.$$

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H_0 : $\beta_1 = \beta_2 = \dots = \beta_k = 0$ (k restrictions) against

H_1 : $\beta_j \neq 0$ for some $j = 1, \dots, k$.

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- ▶ The restricted model is given by

$$Y_i = \beta_0 + U_i,$$

and since $\hat{\beta}_0 = \bar{Y}$ in this model,

$$SSR_r = \sum_{i=1}^n (Y_i - \bar{Y})^2 = SST \text{ and } SSR_{ur} = SSR.$$

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Test of model significance

- ▶ The F statistic for the model significance test and its p -value is reported by Stata as in the top part of the regression output.

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Total	148.329751	525	.28253286

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